



Pitches	50.6%	14.2%	23.3%	57.1%
n = 403	Swing%	Whiff%	CSW%	Zone%

KEY FINDING

Low HAA = 3× more whiffs than High HAA

Low HAA (≈1.08°) generates 25.9% Whiff% — 3× the rate of High HAA (8.5%).

When Beau’s fastball exits the hand at a lower horizontal approach

angle, it arrives from the 1B side and finishes outside to RHH.

That deceptive angle makes it extremely difficult to square up,

driving the dramatic difference in swing-and-miss across folds.

HAA and PlateLocSide share a 0.90 correlation — the approach

angle dictates where the ball ends up, not the other way around.

Low HAA (25.9% Whiff) Mid HAA (11.4% Whiff) High HAA (8.5% Whiff)

TARGET RELEASE POINT

Current RelSide: -2.35 ft

Target RelSide: -2.25 ft

Delta: Δ = +0.10 ft (≈1.2 in toward 1B/glove side)

Expected outcome: Whiff% → ~25.9% (from 14.2% overall)

For a LHP, shifting release toward the 1B / glove side

lowers HAA → ball arrives from 1B side → stays outside

vs RHH — the most effective approach angle.

Guardrail: Zone% drops to 41.5% in Low-HAA fold.

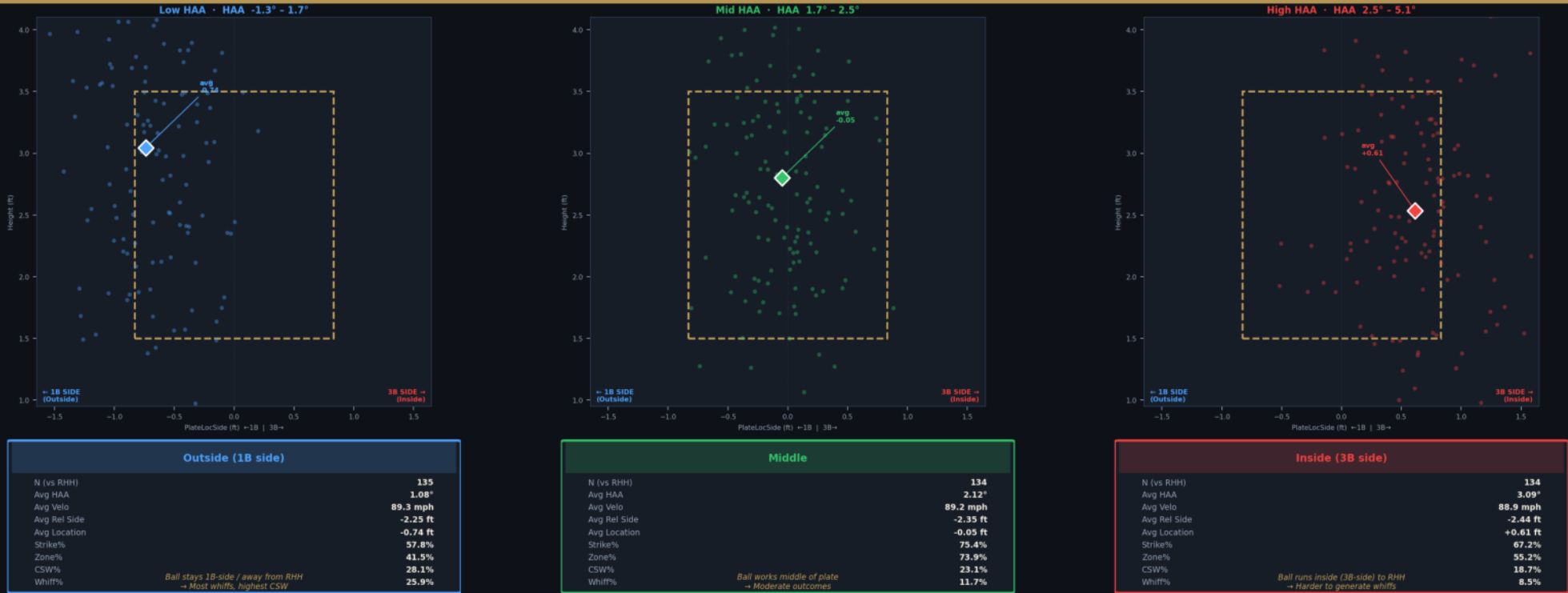
Pair with location discipline or pitch-to-contact mix.

Beau Elson · HAA Fold Analysis



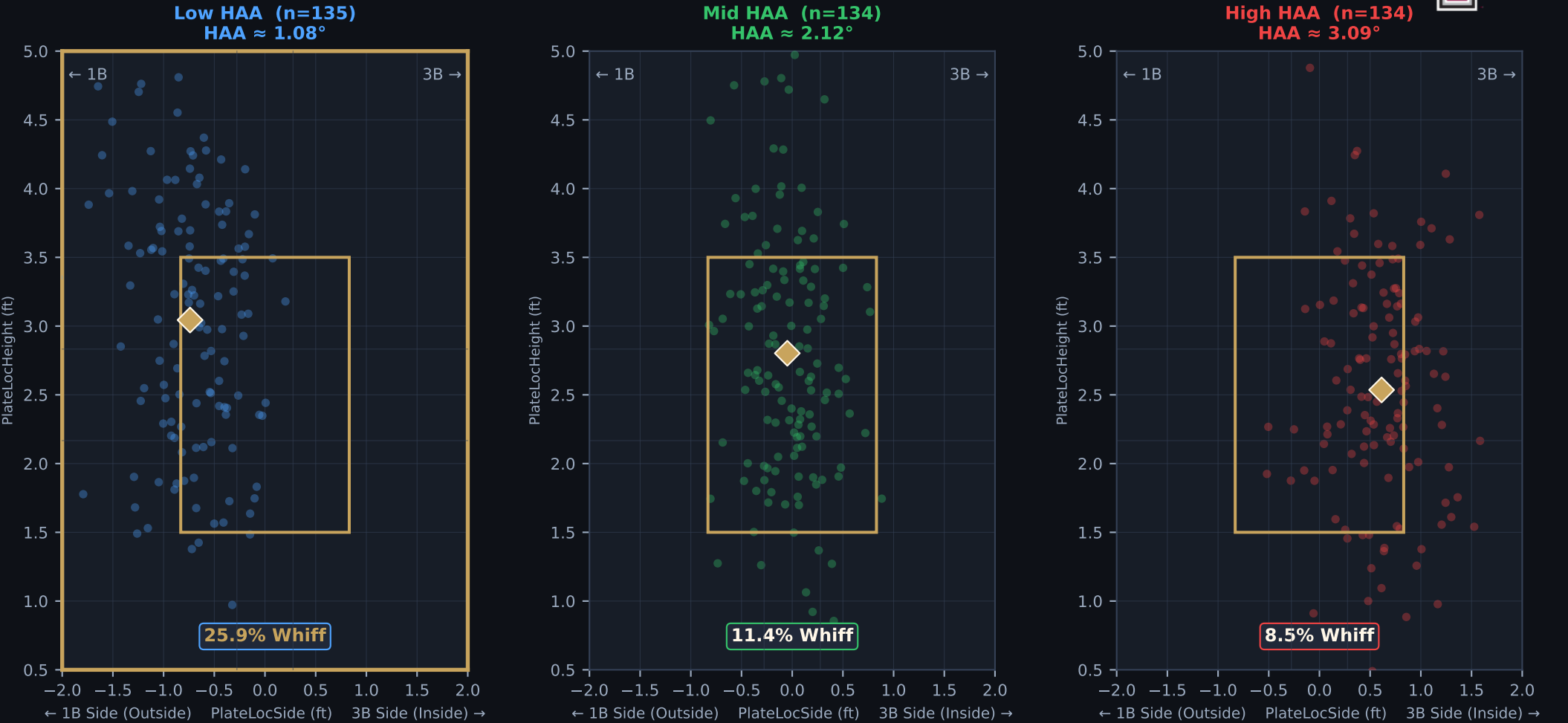
Elson, Beau (LHP) — Fastball HAA Fold Analysis · vs RHH Only

Fastballs split into 3 equal HAA terciles · Positive HAA = ball arriving from 3B side (inside to RHH) · Negative/low HAA = ball arrives from 1B side (outside to RHH) · Low HAA most effective



Three HAA tercile folds (Low / Mid / High) vs RHH | Low fold: HAA ≈ 1.08° → 25.9% Whiff% · Mid fold: HAA ≈ 2.12° → 11.4% Whiff% · High fold: HAA ≈ 3.09° → 8.5% Whiff%

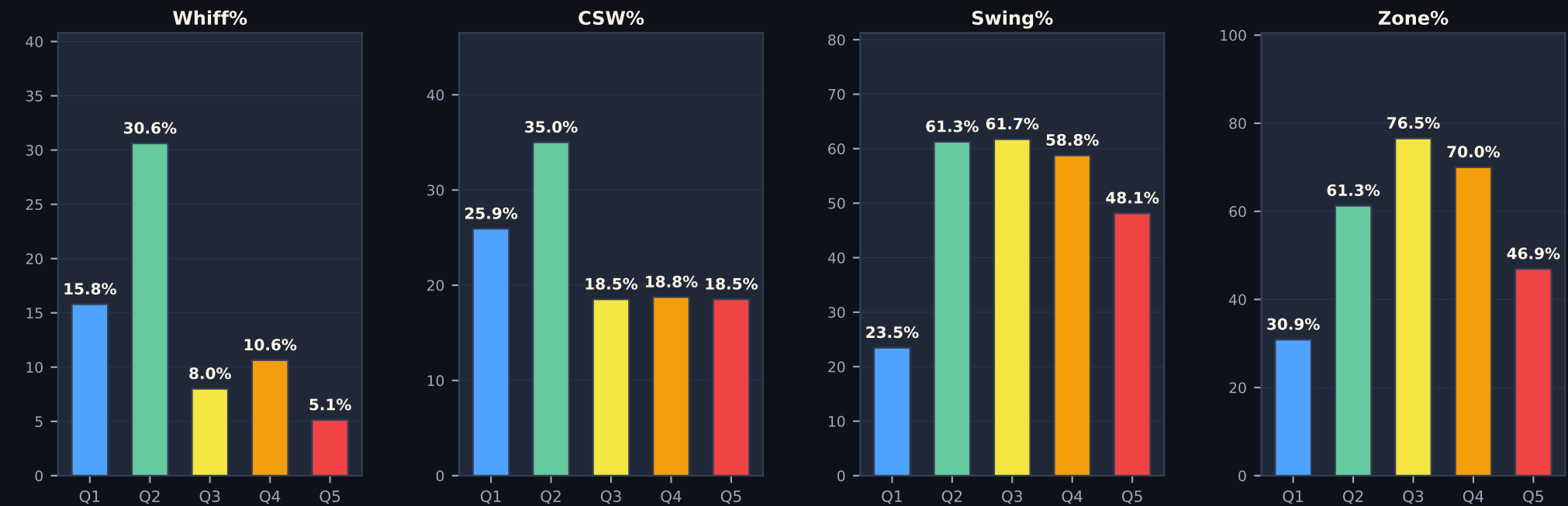
Beau Elson · Zone Location by HAA Fold (vs RHH)



Fold Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Low	135	1.08°	-2.25	5.66	40.0%	25.9%	28.1%	41.5%	35.7%	64.3%	22.8%	83.3%	55.6%
Mid	134	2.12°	-2.35	5.61	59.0%	11.4%	23.1%	73.9%	26.3%	73.7%	17.1%	90.4%	66.7%
High	134	3.09°	-2.44	5.58	53.0%	8.5%	18.7%	56.0%	24.0%	76.0%	23.7%	94.7%	78.6%

	1B Side (Outside)	Middle	3B Side (Inside)
High	33.3%	30.3%	12.0%
Mid	11.8%	3.6%	0.0%
Low	8.3%	6.2%	8.3%

Beau Elson · 5-Fold HAA Analysis (Quintiles vs RHH)



5-Fold (Quintile) Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Q1 Low	81	0.78°	-2.25	5.68	23.5%	15.8%	25.9%	30.9%	52.0%	48.0%	12.5%	91.7%	71.4%
Q2	80	1.62°	-2.28	5.63	61.3%	30.6%	35.0%	61.3%	30.6%	69.4%	48.4%	79.4%	46.7%
Q3	81	2.12°	-2.35	5.61	61.7%	8.0%	18.5%	76.5%	21.0%	79.0%	5.3%	91.8%	100.0%
Q4	80	2.58°	-2.38	5.60	58.8%	10.6%	18.8%	70.0%	23.2%	76.8%	16.7%	90.7%	75.0%
Q5 High	81	3.37°	-2.47	5.57	48.1%	5.1%	18.5%	46.9%	26.3%	73.7%	25.6%	100.0%	81.8%

Q1 (lowest HAA, avg 0.78°) → outside to RHH Q5 (highest HAA, avg 3.37°) → inside to RHH